



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. EXAMINATIONS - FEBRUARY 2023
SUBJECT: MAT 1171 / MAT-1171 - ENGINEERING MATHEMATICS - I
(MAKEUP)

ENGINEERING MATHEMATICS - I [MAT 1171-CHM]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) For the given data construct the interpolating polynomial using forward difference formula, and hence find $f(0.5)$.

x	-2	-1	0	1	2	3
f(x)	15	5	1	3	11	25

(4)

- 1B) Solve $(x^3y^2 + xy)dx = dy$ (3)

- 1C) Evaluate $\int_0^1 \sqrt{1+x^4} dx$ (3)
using Simpson's 1/3-rule, with $h = 1/6$.

- 2A) Using Runge-Kutta Method of order 4, find $y(0.4)$, given $y' = -2xy^2$, $y(0) = 1$ with $h = 0.2$. (4)

- 2B) Solve: $(\cos(x+y) - 2xy^2) dx + (\cos(x+y) - 2x^2y) dy = 0$ (3)

- 2C) Compute the value of x , when $y=8$ by Lagrange's inverse interpolation formula

x:	-2	-2	1	2
y:	-7	2	0	11

(3)

- 3A) Using Gauss elimination method, solve the following system of linear equations:
 $2x + y + 4z = 12$; $4x + 11y - z = 33$; $8x - 3y + 2z = 20$. (4)

- 3B) Solve $(D^2 - 6D + 25)y = \sin x$ (3)

- 3C) Using Newton-Raphson method, find a root of the equation $x \sin x + \cos x = 0$, correct to four decimal places (take $x_0 = \pi$) (3)
Take x in radians.

- 4A) (5)

Using Gram Schmidt orthogonalization, construct an orthonormal basis from the following set of vectors $B = \{(1,1,1), (1,0,1), (1,1,-1)\}$. for $\mathbb{R}^3$.

- 4B) Find all the eigenvalues of the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ and find the eigenvector corresponding to any one of the eigenvalues. (5)

- 5A) Using Taylor series method, find $y(0.1)$ and $y(0.2)$ to four decimal places given $\frac{dy}{dx} + 1 = x^2y$, $y(0) = 1$. (5)

- 5B) Using row reduced echelon form, find the rank of the matrix $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 1 & 4 & 3 \end{pmatrix}$ (5)

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